

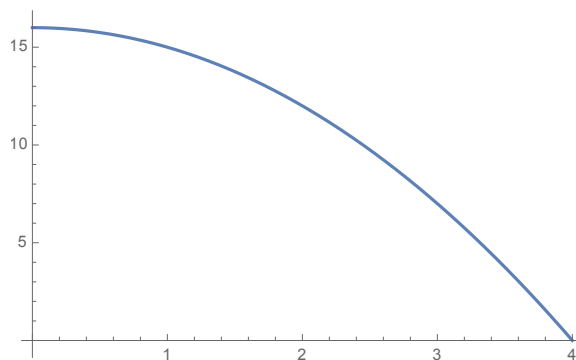
## 5.2 The Definite Integral

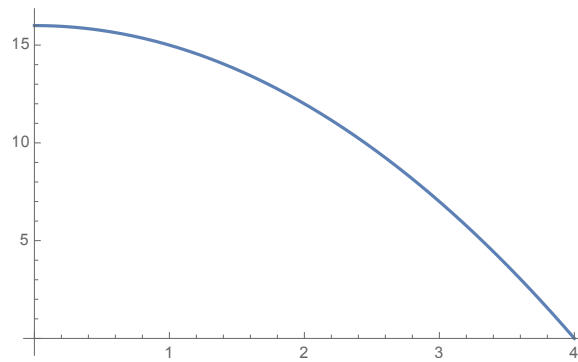
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Many of the formulas used to calculate values in the upcoming sections (e.g. volume, work, center of mass, probability) will be developed by first building Riemann sums. While you were introduced to Riemann sums in your previous calculus course, you may not have realized their importance. Since Riemann sums play such a significant role in this course, we will spend some time reviewing.

### Motivation:

Suppose we wish to calculate the area of a shaded region  $R$  that lies above the  $x$ -axis, below the graph of  $f(x) = -x^2 + 16$ , and between the vertical lines  $x = 0$  and  $x = 4$ . How might we estimate this area?





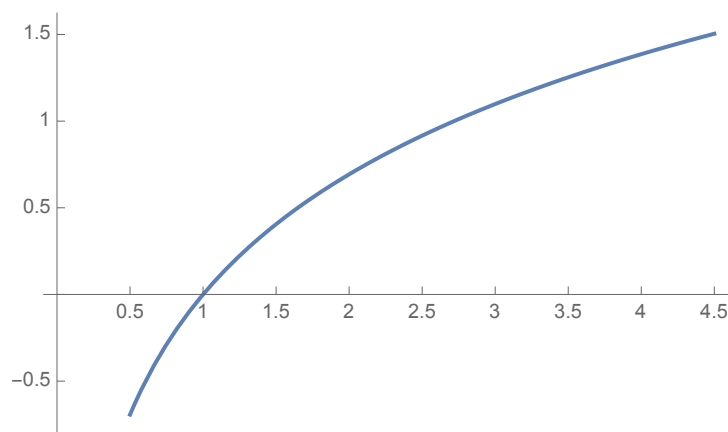
What is the area formula for a rectangle? Why does this make rectangles a convenient shape to use in order to determine the area bounded by a curve and the  $x$ -axis?

How does the number of rectangles that we use affect the accuracy of our measurement? How do we determine how many rectangles we should use to estimate area?

How do we know whether our approximation using rectangles is an overestimate or underestimate?

## Example 1

Suppose you want to find the *signed* area of the region  $R$  that lies between the  $x$ -axis, the graph of  $f(x) = \ln(x)$ , and between the vertical lines  $x = 0.5$  and  $x = 4.5$ .

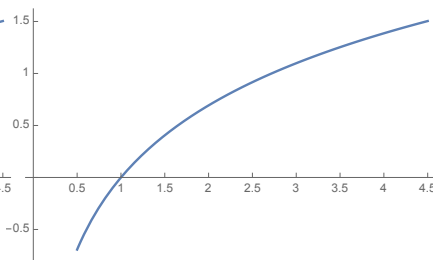
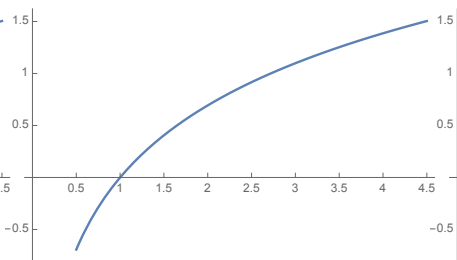
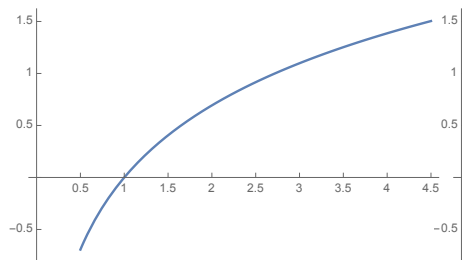


Approximate the signed area to two decimal places with 4 rectangles using

1. right hand sum

2. left hand sum

3. mid-point rule

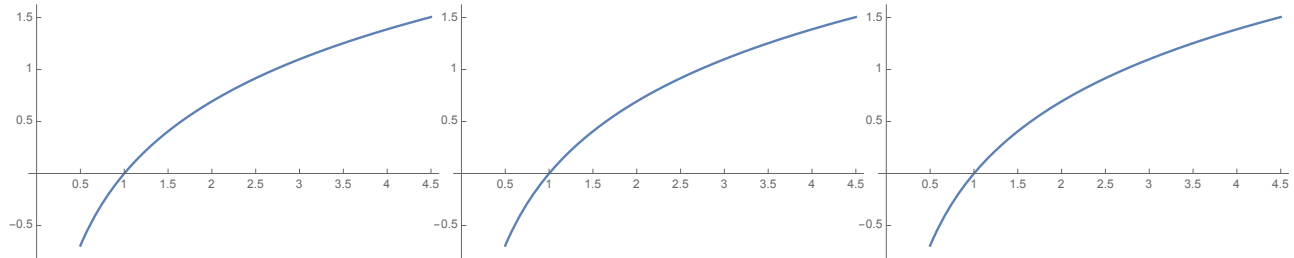


Approximate the signed area to two decimal places with 4 rectangles using

1. right hand sum

2. left hand sum

3. mid-point rule



Consider Example 1 on the previous page.

1. What can we conclude about the actual value of the signed area? Why?
2. Do the values calculated estimate the true area under the curve?  
Why or why not?
3. How could we get a more accurate estimation of the signed area?

### Definition

Let  $f$  be a function defined on the interval  $[a, b]$ . We can divide the interval  $[a, b]$  into  $n$  sub-intervals of equal width

$$\Delta x =$$

Let  $x_0 = a, x_1, x_2, \dots, x_{n-1}, x_n = b$  be the endpoints of these sub-intervals. Let  $x_1^*, x_2^*, \dots, x_n^*$  be any sample points in these sub-intervals such that  $x_i^*$  is in the sub-interval  $[x_{i-1}, x_i]$ . Then the **definite integral of  $f$  from  $a$  to  $b$**  is

$$\int_a^b f(x) \, dx =$$

What is the definite integral?

Why does this definition make sense?

What is a reasonable point to choose for  $x_i^*$ ?

**Example 2**

Write  $\int_1^3 \sqrt[3]{x^2 + 2} \, dx$  as a Riemann sum.



**Example 3**

Interpret  $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2 \left(3 + \frac{3i}{n}\right)}{1 + \left(3 + \frac{3i}{n}\right)^2} \left(\frac{3}{n}\right)$  as a definite integral.

## Riemann Sums Recap

1. What is the relationship between a Riemann sum and a definite integral?
2. Which is more accurate? A Riemann sum with  $n$  sub-intervals or the definite integral?

3. How can we make a Riemann sum more accurate?

4. When might we use Riemann sums rather than the definite integral?

## Properties of the Integral

Let  $f$  and  $g$  be continuous functions. Let  $c$  be a constant.

- $\int_b^a f(x) \, dx = - \int_a^b f(x) \, dx$
- $\int_a^a f(x) \, dx = 0$
- $\int_a^b c \, dx = c(b - a)$
- $\int_a^b (f(x) \pm g(x)) \, dx = \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx$
- $\int_a^b cf(x) \, dx = c \int_a^b f(x) \, dx$
- $\int_a^b f(x) \, dx + \int_b^c f(x) \, dx = \int_a^c f(x) \, dx$
- If  $f(x) \geq 0$  for all  $a \leq x \leq b$ , then  $\int_a^b f(x) \, dx \geq 0$ .
- If  $f(x) \geq g(x)$  for all  $a \leq x \leq b$ , then  $\int_a^b f(x) \, dx \geq \int_a^b g(x) \, dx$ .
- If  $m \leq f(x) \leq M$  for all  $a \leq x \leq b$ , then

$$m(b - a) \leq \int_a^b f(x) \, dx \leq M(b - a).$$